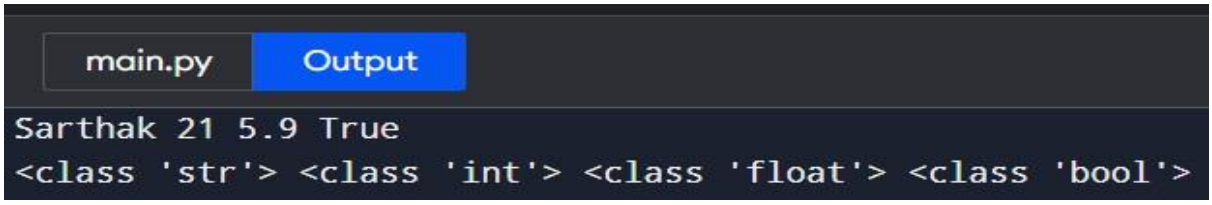
Experiment no:-1

Title:- write the program to demonstrate variables ,data types ,operators and type casting in python

1. Create variables to store your name, age, height, and whether you are a student. Print all variables and their data types.

   name =
   "Sarthak" age =
   21 height = 5.9
   is_student = True

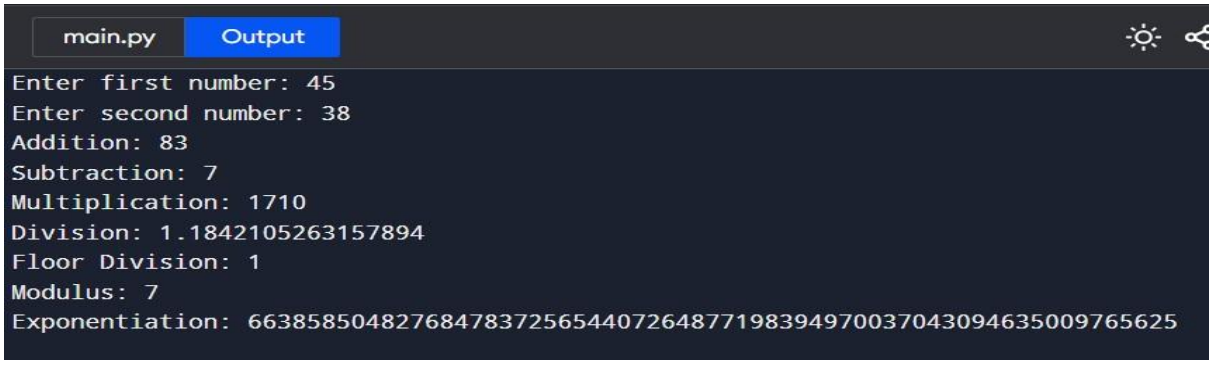
   print(name, age, height, is_student)
   print(type(name), type(age), type(height), type(is_student))



2. Take two integer inputs from the user and perform addition, subtraction, multiplication, division, floor division, modulus, and exponentiation. Print the results

   a = int(input("Enter first number: "))
   b = int(input("Enter second number: "))

   print("Addition:", a + b)
   print("Subtraction:", a - b)
   print("Multiplication:", a * b)
   print("Division:", a / b)
   print("Floor Division:", a // b)
   print("Modulus:", a % b)
   print("Exponentiation:", a ** b)



. 3. Take two numbers as input and compare them using all comparison operators (>, <, ==, !=, >=, <=). Print the boolean results

main.py Share Run Output

```
1  # Program 3
2  x = float(input("Enter first number: "))
3  y = float(input("Enter second number: "))
4
5  print(x > y, x < y, x == y, x != y, x >= y, x <= y)
6
```

```
Enter first number: 4
Enter second number: 5
False True False True False True

=== Code Execution Successful ===
```

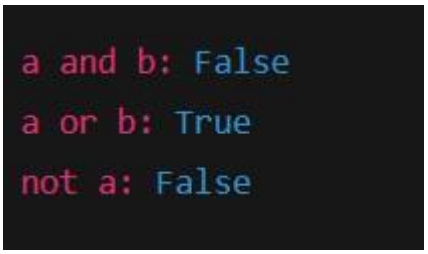
4) Check logical operators: Create two boolean variables and apply and, or, and not operators. Print the Result a =

True b = False

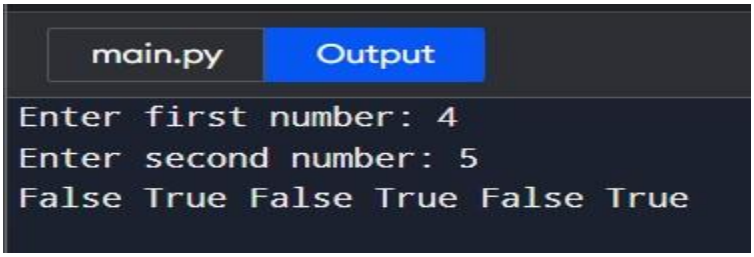
print("a and b:", a and b)

print("a or b:", a or b)

print("not a:", not a)



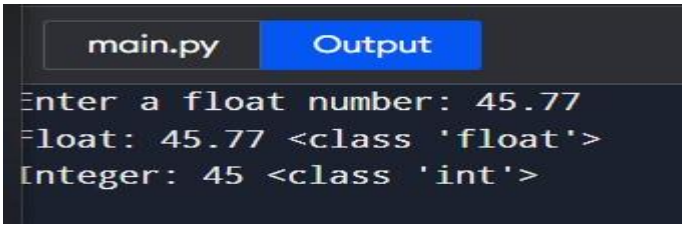
5. Type casting from string to integer and float: Take a numeric string from the user, convert it to integer and float, and print their types. num_str = input("Enter a number as string: ") num_int = int(num_str) num_float = float(num_str) print(num_int, type(num_int)) print(num_float, type(num_float))



6. Type casting from float to integer: Take a float input from the user and convert it to an integer. Print both values and types. num = float(input("Enter a float number: ")) num_int = int(num)

print("Float:", num, type(num)) print("Integer:",

num_int, type(num_int))

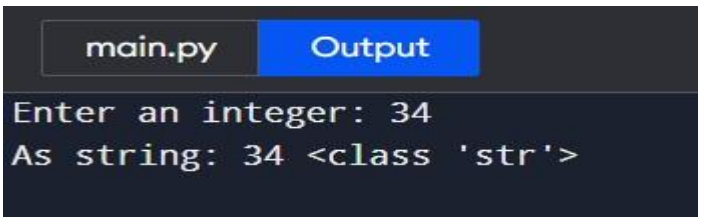


7. Type casting from integer to string: Take an integer input and convert it to a string. Print the result and its type.

```
num = int(input("Enter an integer: "))

num_str = str(num) print("As string:",

num_str, type(num_str))
```



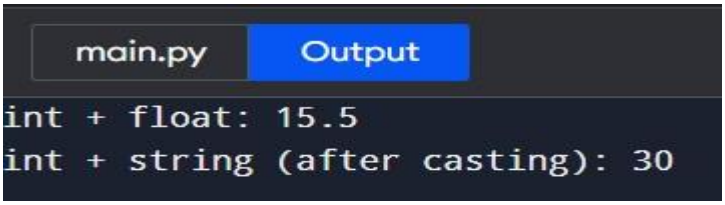
8. Perform arithmetic operations on variables of different data types (int + float, int + string after casting, etc.) and print results.

```
x = 10 y = 5.5 z = "20" result1 = x + y

result2 = x + int(z) print("int + float:",

result1) print("int + string (after

casting):", result2)
```



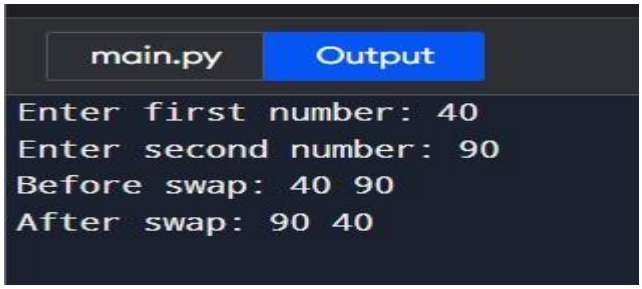
9. Swap two numbers using a temporary variable and print the result before and after swapping.

```
a = int(input("Enter first number: ")) b

= int(input("Enter second number: "))

print("Before swap:", a, b) temp = a a =

b b = temp print("After swap:", a, b)
```



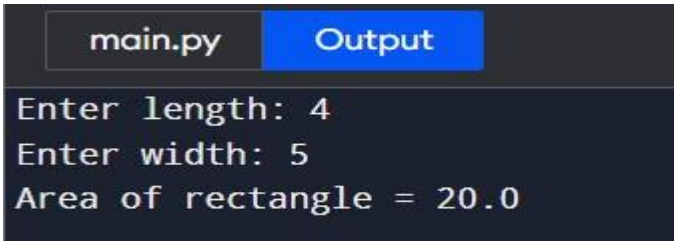
10. Use input to calculate the area of a rectangle: Take length and width as input, calculate area, and print. Ensure inputs are properly converted to float.

```
length = float(input("Enter length: "))

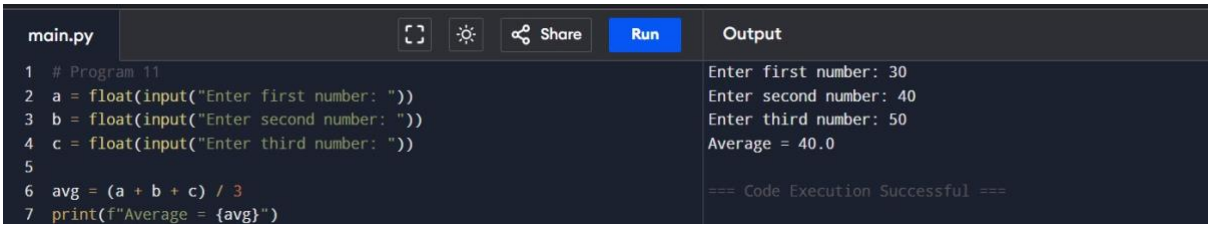
width = float(input("Enter width: "))
```

area = length * width print(f"Area of

rectangle = {area}")

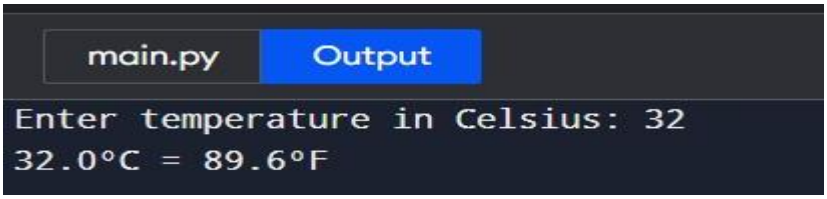


11. Calculate the average of three numbers entered by the user. Use type casting if necessary.

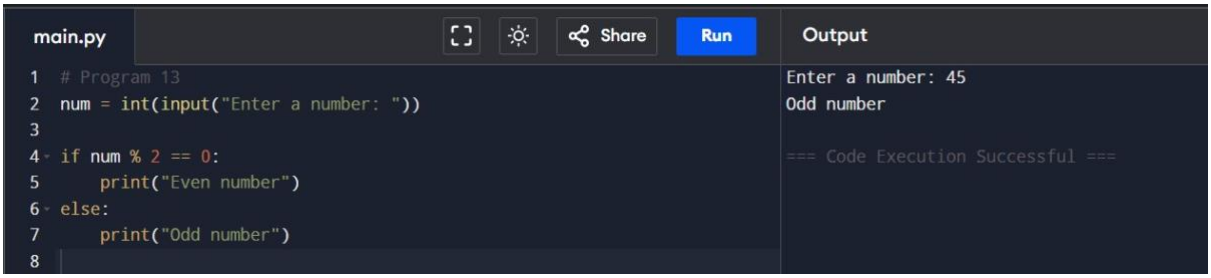


12. Convert temperature: Take temperature in Celsius as input and convert it to Fahrenheit. Print both values. c

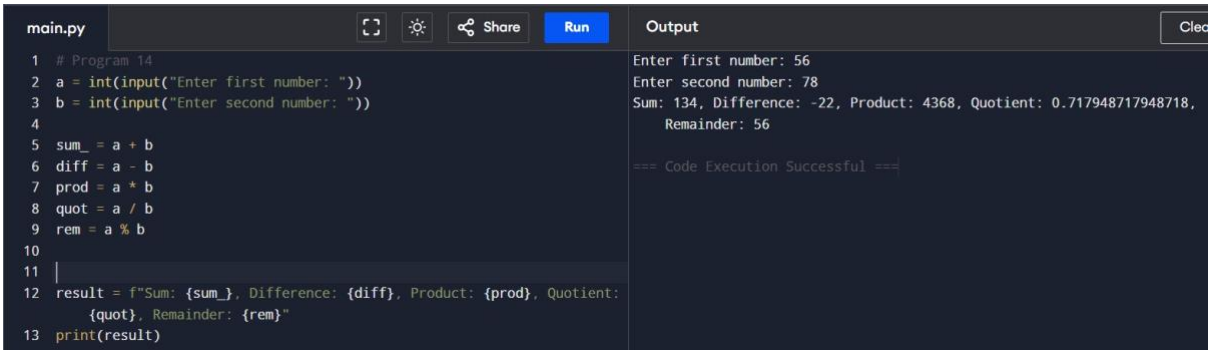
= float(input("Enter temperature in Celsius: ")) f = (c * 9/5) + 3232 print(f"{c}°C = {f}°F")



13. Check if a number is even or odd using the modulus operator and print the result.



14. Perform a series of calculations: Take two numbers, calculate sum, difference, product, quotient, and remainder. Then cast all results to strings and print them concatenated in a single sentence.



15. User input for personal details: Take name, age, and height as input, cast age name = input("Enter your name: ") age = int(input("Enter your age: ")) height = float(input("Enter your height in feet: ")) print(f"Name: {name}, Age: {age}, Height: {height} ft")

